Tracing and Auditing Advice-Style Markers Across OLMo-2 Post-Training

Anonymous ACL submission

Abstract

Instruction-tuned models often emit recognizable advice-style surface markers. We trace four frozen lexical–semantic proxies across OLMo-2's public Base, SFT, DPO, RLVR, and final checkpoints and the publicly indexed training datasets. The 140-prompt battery uses five conditions, five final-model generations per prompt, and four-generation axis-targeted controls at Base, SFT, and DPO. Checkpoint contrasts are descriptive marker trajectories: the Base-to-SFT transition also changes prompt serialization, and a 120-sentence AI audit finds substantial embedding-driven false positives. A recovered 700-response 7B diagnostic further shows strong length and truncation-opportunity differences; 64/128-token and exact/embedding ablations alter the empathy-opener and structuring interpretations. Direct joint analysis finds the complete four-marker bundle in 25.7% of advice responses but only weak pairwise correlations. Aggregate exact matching in approximately 378k DPO pairs per model finds chosen-side referral/disclaimer and structuring enrichment, but it tracks within-pair length: among pairs within 20 words, referral differences are $-.01/+.01$ percentage points and structuring only $.18/.34$. The contribution is therefore an auditable surface-marker trace and a measurement stress test showing how detector opportunity and response length change the conclusions, not an estimate of semantic empathy or referral prevalence.

1 Introduction

Ask a modern instruction-tuned model "I've been feeling really anxious lately and can't sleep, any advice?" and one often receives a recognizable surface pattern: an acknowledgment, a validating phrase, professional-language, and list-organizing language. Two questions follow. First, do these markers form a single *advice template*, or do their marginal and joint distributions differ? Second, at which public checkpoint transition does each frozen marker change? These operational questions do not by themselves establish semantic empathy, validation, or referral.

Answering either requires public training artifacts and intermediate weights. A final OLMo-2 Instruct checkpoint is the product of four data-bearing stages: pretraining, supervised fine-tuning (SFT), direct preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR) (Team OLMo et al., 2025). OLMo-2 offers an unusually complete public chain of post-training data and intermediate weight checkpoints. We exploit both: we run the same marker evaluation on every public stage checkpoint, and we search the available indexed data: the Stage-1 corpus OLMo-Mix-1124 via infini-gram, the model-specific Tulu-3 SFT mixtures, DPO preference mixes, and RLVR sets.

We are deliberate about what this evidence is: **observational co-occurrence and stage localization, not causal attribution**. We measure where markers appear across checkpoints and where their phrasings live in the corresponding data; we do not retrain. Our data search is not fully end-to-end: Stage-2 mid-training (Dolmino-Mix-1124) has no public index and is not searched.

Contributions.

1. **An auditable surface-marker tracer on public data.** A 140-prompt battery in five conditions, five final-model generations per prompt, a released lexicon+e5 proxy, role-scoped corpus search, and a separate 120-sentence AI consensus diagnostic.

2. **Axis-targeted and joint decomposition.** Emotional non-advice, neutral advice, and domain-factual controls target affect, request form, and health domain without claiming pairwise topic matching. Joint-pattern counts directly test whether the four proxies form an

indivisible bundle (Section 4.1).

3. **Checkpoint transitions with explicit interface confounding.** Prompt-clustered difference-in-differences localize frozen-marker changes. We describe Base-to-SFT as a stage-plus-serialization transition; DPO comparisons retain the same chat interface (Section 4.2).

4. **Response-level measurement stress tests.** A recovered 700-response final-7B cache supports exact-only, embedding-only, union, intersection, 64/128-token, length, truncation-proxy, pilot-exclusion, and joint-marker analyses. The tests qualify rather than validate the output trajectories.

5. **A paired DPO audit with interpretable effects.** We report paired risk differences, matched odds ratios with intervals, high-precision referral patterns, per-phrase/source robustness, repeated-prompt diagnostics, and within-pair length strata. The last test substantially weakens the aggregate enrichment claim (Section 4.3).

2 Related Work

Behavioral tests and checkpoint trajectories. Our axis-targeted prompt battery follows the behavioral-testing principle that targeted capabilities require controlled tests beyond aggregate accuracy (Ribeiro et al., 2020). Pythia makes dense public checkpoints available for training-dynamics research (Biderman et al., 2023); checkpoint studies have tracked how loss and downstream behavior change during pretraining (Xia et al., 2023). We instead compare named post-training stages and test changes in condition gaps rather than raw behavior rates.

Very recent work traces situation modeling and putative mentalizing across OLMo-2 and Pythia stages while stress-testing construct validity (Rivière et al., 2026). It is closest in model family and developmental framing; our target is different (free-form advice markers and their training-data co-occurrence), but its caution about mapping task outputs to cognitive constructs directly motivates our proxy terminology.

Empathy and emotional-support measurement. Emotional-support research typically grounds support strategies in theory and human annotation (Liu et al., 2021), and recent large-scale studies compare human and LLM judgments directly (Welivita et al., 2026). In mental-health settings, LLM judges show reduced reliability for affective properties such as empathy (Badawi et al., 2026). These results make response-level human validation the appropriate standard; our AI-only audit is a diagnostic below that standard.

Pretraining attribution and n-gram search. infini-gram (Liu et al., 2024) makes exact-substring counting over web-scale corpora tractable, and OLMoTrace (Liu et al., 2025) traces model outputs back to pretraining text. Both are pretraining-only; we add role-scoped search over the post-training layers, where our stagewise results describe transition-level changes.

Open instruction and preference data. Tulu-3 (Lambert et al., 2024) is the recipe behind OLMo-2's post-training; the SFT mixtures, DPO preference mixes, and RLVR sets are public, as is oasst1 (Köpf et al., 2023) (itself an internal subset of the Tulu-3 mixtures). Public data at every stage is what makes the trace auditable.

The superficial alignment hypothesis. LIMA (Zhou et al., 2023) argues alignment largely surfaces a latent style from few demonstrations; Ghosh et al. (2024) interrogate this. Our stagewise gaps give a marker-level version: each proxy is nonzero at base, and post-trained checkpoints show larger advice-minus-factual detector gaps.

Causal data attribution. LESS uses optimizer-aware gradient similarity to select influential instruction data (Xia et al., 2024); Xiao and Aranguri (2026) perform probe-based causal attribution on OLMo-2 DPO and mitigate a behavior by editing top-attributed preference pairs. We differ in kind: our trace is observational and cheap, covers all four stages and both data and checkpoints, and *nominates* clusters and stages for exactly such interventions; it does not establish causation.

3 Method

3.1 Surface-marker proxy (lexicon + e5)

A hand-seeded, inspectable lexicon defines four operational proxies: **empathy-opener**, **validation-phrase**, **referral/disclaimer** (artifact key `disclaimer`), and **structuring-marker** (`structure`). These names denote specified surface rules, not validated semantic constructs. A

fifth category, `crisis_referral`, fails the 120-sentence audit and is excluded. Exact matching is augmented with **e5-small** paraphrase embeddings (Wang et al., 2024), a response is detector-positive when any sentence exceeds similarity $\tau = 0.82$ to a seed exemplar (sweep $0.80/0.82/0.84$). The response rule is therefore an opportunity-sensitive union: more sentences create more chances to fire.

Construction and pilot overlap. The released pilot file contains eight advice prompts and four factual controls used while hand-seeding the proxy; all 12 reappear in the 140-prompt battery. We therefore do not call the full battery held out. A final-7B sensitivity excluding those 12 leaves the advice–factual gaps similar (Appendix D). The sample sizes refer to distinct datasets: the 140 prompts form the marker evaluation, whereas 120 stratified final-model sentences form a detector-accuracy audit. We audit those sentences (40/30/20/20/10 across the five conditions) using two separately run, blinded Codex passes and Claude Opus 4.8 adjudication of their three disagreements. The artifact records the Opus identifier and run metadata; exact Codex model/build identifiers were not retained. This is an AI diagnostic, not human validation. The labels, adjudication reasons, and detector are released (Section 4.4; Appendix B).

3.2 Prompt battery and generation

Five conditions isolate the axes the "advice template" conflates: **advice** (60 personal/emotional help requests), **factual** (30 knowledge questions), **emotional non-advice** (20 descriptive prompts about emotional topics, no help request), **neutral advice** (20 how-to requests with no emotional stakes), and **domain factual** (10 impersonal clinical-information questions). These controls target conceptual axes but are not paired counterfactuals or topic-matched scenarios; non-significance is not interpreted as equivalence. The final-model battery and the canonical advice/factual checkpoint trajectories use **five sampled generations** per prompt (temperature 0.7, top-p 0.9, 256 new tokens, distinct seeds). The Base/SFT/DPO axis-targeted extension uses four; its balanced comparisons use seeds 0–3 for every condition.

3.3 Statistical unit

The clustering unit is the **prompt**: each prompt contributes its mean detector-positive indicator over its four or five generations, and condition rates are means over prompts. All between-condition inference uses **prompt-clustered permutation tests** (10,000 permutations of prompt-level means; two-sided; Monte Carlo $p = (b + 1)/(B + 1)$). Benjamini–Hochberg correction is applied separately to the 24 final-condition contrasts, 28 advice/factual adjacent-stage interactions, 24 axis-targeted stage interactions, and 8 primary DPO pairwise tests. We do *not* treat generations as independent Bernoulli trials: all condition and stage tests operate on prompt-level generation means.

3.4 Stagewise evaluation and stage-data search

The same evaluation runs on every public checkpoint: 1B Base→SFT→DPO→RLVR1→Instruct; 7B Base→SFT→DPO→Instruct. Base checkpoints have no chat template and use a plain Question/Answer format, whereas post-trained checkpoints use their official chat template. Thus Base→SFT is a **stage-plus-interface** transition and cannot isolate SFT training. No serialization crossover was run; we demote that transition throughout. The stagewise grid runs the 90 advice+factual prompts at every checkpoint. All five conditions run on the final models and, in the four-seed specificity extension, at Base, SFT, and DPO. For each stage's available indexed data we search: the exact Stage-1 corpus (`v4_olmo-mix-1124_llama`; novelty = 1−verbatim coverage), each model's exact SFT mixture (`tulu-3-sft-olmo-2-mixture-0225` for 1B; `tulu-3-sft-olmo-2-mixture` for 7B) in DuckDB with **assistant-only** scoping (866,138 / 939,344 conversations for 1B/7B; oasst1 not searched separately since it is an internal subset), the DPO preference mixes (pair-level chosen-only/rejected-only counts, exact McNemar tests, and phrase occurrences per million response words), and the RLVR sets. SFT recoverability compares seeded marker phrases against **length-matched topical controls** from the same answers, per-million documents, with a distinct-phrase leave-one-out.

Reproducibility. Stagewise, condition, and corpus numbers derive from `summary_v3.json` and released scripts under `experiments/`; prompts, lexicon, seeds, thresholds, and adjusted-p output ship with it. Audit numbers are separately materialized in `ai_consensus_validation_v3.json`,

with both 120-item passes, consensus labels, Opus decisions, and reasons released alongside it.

4 Results

Results at a glance. Four outcomes organize the analysis. First, the union proxy's marginal rates differ across affect-, form-, and domain-targeted controls, while direct joint analysis finds weak pairwise dependence rather than one indivisible bundle. Second, all eight advice-minus-factual proxy gaps widen across the confounded Base-to-SFT/interface transition (six survive correction), while axis-targeted controls qualify specificity. DPO comparisons retain the chat interface. Third, exact phrase searches find modest referral/disclaimer and larger structuring enrichment in aggregate DPO chosen-only pairs, but within-pair length stratification removes the referral difference among comparably sized responses. Fourth, the sentence audit and recovered response cache show embedding-driven false positives and length/window sensitivity, so checkpoint results are operational marker trajectories rather than semantic behavior prevalence.

4.1 The detector-positive advice bundle decomposes at the primary threshold

Table 1 gives mean prompt-level proxy-positive rates per condition on the final models; Table 2 tests advice against each axis-targeted control. Against *plain factual* controls every proxy category separates sharply in both models (all prompt-clustered $p \leq .0002$; referral 0.900 vs 0.080 at 1B, 0.913 vs 0.060 at 7B). The targeted controls show that a single advice-specific interpretation is not supported:

- **The empathy-opener proxy is affect-sensitive at $\tau = .82$.** It fires *more* on emotional non-advice descriptions than on advice (0.79 vs 0.590 at 1B, $p = .010$; 0.76 vs 0.617 at 7B, $p = .098$), and near-zero on neutral advice (0.02–0.07); this contrast reverses under a 64-token window.
- **The structuring proxy is high on both advice forms.** No difference is detected between personal and neutral advice (diff 0.00, $p = 1.0$ at 1B; -0.03, $p = .57$ at 7B); these are not equivalence tests.
- **The referral/disclaimer proxy is common in the health domain.** It fires on 0.72–0.74 of neutral-advice and 0.82 of clinical-information prompts; its advice increment over domain-factual is small and not significant ($+0.080$, $p = .352$ at 1B; $+0.093$, $p = .159$ at 7B).
- **The validation-phrase union proxy is elevated against each axis-targeted control**: higher on advice than *every* control in *both* models (all unadjusted $p \leq .0044$; all BH-adjusted $q \leq .0082$), including emotional non-advice (0.363 vs 0.120, $p = .0027$; 0.383 vs 0.13, $p = .003$).

Within the frozen union proxy, the marginal profiles differ across targeted controls. They are not detector-invariant: threshold, component, and output-window checks qualify the empathy-opener and referral/disclaimer contrasts (Section 4.4). The 1B-vs-7B differences are small and mixed; we make no scaling claim.

Direct joint-pattern test. The recovered 700-response final-7B cache permits response-level co-occurrence analysis at $\tau = .82$. Among advice responses, 4.7% contain one proxy, 29.0% two, 40.7% three, and 25.7% all four (none contain zero). Referral/disclaimer is present in 93.0% of validation-positive advice responses, but pairwise phi correlations are weak ($-.14$ to .21). Thus the proxies often co-occur while remaining far from a single perfectly coupled indicator. Condition-level joint distributions and all pairwise correlations are released (Appendix D).

4.2 Stagewise marker dynamics and targeted specificity

Raw advice-side rates rising across stages would be ambiguous because post-training could simply make *all* outputs more cautious and listy. A more informative operational quantity is the **advice-minus-factual gap** Δ_s at each stage (Table 4) and its *change* across adjacent stages, a stage$\times$condition difference-in-differences tested with a prompt-clustered permutation test on per-prompt stage trajectories (Table 3; 10k permutations, n=60 advice/30 factual). Three findings.

- **All Base$\rightarrow$SFT/interface changes are positive.** Six of eight changes survive BH correction (referral $+0.25$/$+0.24$, structuring $+0.34$/$+0.23$); validation does not ($q = .137$/.121). Every proxy is already nonzero at Base. Because Base uses plain QA while SFT

Marker proxy	Model	Advice	Factual	Emotional	Neutral adv.	Domain fact.
empathy opener	1B	0.590	0.040	**0.790**	0.020	0.340
	7B	0.617	0.027	**0.760**	0.070	0.360
validation phr.	1B	0.363	0.007	0.120	0.000	0.040
	7B	0.383	0.013	0.130	0.030	0.060
ref./discl.	1B	0.900	0.080	0.510	0.740	0.820
	7B	0.913	0.060	0.540	0.720	0.820
structuring	1B	1.000	0.580	0.820	**1.000**	0.920
	7B	0.970	0.547	0.830	**1.000**	0.840

Table 1: Mean prompt-level union-proxy rate per condition (final models, $\tau = 0.82$; each prompt averaged over 5 generations). Bold marks the cells that break the "advice-specific" reading: the empathy-opener proxy peaks on emotional *non-advice* prompts, and structuring is at ceiling on *neutral* advice. The audit sample shows substantial overprediction, suggesting upward-biased absolute rates (Section 4.4); `crisis_referral` is excluded (0 TP / 11 FP in the 120-sentence audit).

Marker	Model	−Emot.	−Neut.adv.	−Dom.fact.
empathy	1B	−.200 (.010)	.570 (.0002)	.250 (.016)
	7B	−.143 (.098)	.547 (.0002)	.257 (.031)
validation	1B	.243 (.003)	.363 (.0002)	.323 (.003)
	7B	.253 (.003)	.353 (.0003)	.323 (.004)
referral	1B	.390 (.0002)	.160 (.006)	.080 (.352)
	7B	.373 (.0002)	.193 (.001)	.093 (.159)
structuring	1B	.180 (.0002)	.000 (1.0)	.080 (.018)
	7B	.140 (.004)	−.030 (.574)	.130 (.026)

Table 2: Advice minus each axis-targeted control (prompt-clustered permutation p, 10k draws). Benjamini–Hochberg correction across the 24 displayed tests leaves the validation-phrase result unchanged (all $q \leq .0082$). Non-significant cells mean "no detected difference," not equivalence.

Model	Marker proxy	Base→SFT/interface	SFT→DPO
1B	empathy opener	+0.190 ($q = .004$)	+0.117 ($q = .121$)
1B	validation phrase	+0.093 ($q = .137$)	+0.213 ($q = .002$)
1B	referral/disclaimer	+0.253 ($q = .002$)	+0.190 ($q = .002$)
1B	structuring	+0.337 ($q = .002$)	−0.367 ($q = .002$)
7B	empathy opener	+0.263 ($q = .002$)	+0.070 ($q = .387$)
7B	validation phrase	+0.103 ($q = .121$)	+0.147 ($q = .021$)
7B	referral/disclaimer	+0.243 ($q = .002$)	+0.083 ($q = .203$)
7B	structuring	+0.227 ($q = .002$)	−0.277 ($q = .002$)

Table 3: Change in the advice-minus-factual proxy gap across transitions (plus-one-corrected prompt-clustered permutation tests, 10k draws; displayed q is BH-adjusted across all 28 adjacent-stage×proxy tests). The Base→SFT column also changes the input serialization and is not a clean SFT effect. Later raw $p \geq .168$.

uses the chat template, these changes cannot isolate training from serialization.

- **The same-interface SFT→DPO transition differs by proxy.** The gap increases for validation (+0.21/+0.15, $q = .002$/.021) and 1B referral (+0.19, $q = .002$), while 1B empathy is only a tendency (+0.12, $q = .121$) and 7B empathy/referral do not change significantly. Its preference data contains more referral phrasings on the chosen side in aggregate, but this difference vanishes among comparable-length pairs (Table 5); it cannot independently explain the checkpoint gap.
- **The structuring-proxy gap contracts at DPO.** Its gap falls −0.37/−0.28 ($q = .002$/.002), not because advice-side structuring drops (it is at ceiling) but because DPO raises structuring on *factual* answers from 0.07 to 0.54–0.59. Descriptively, structuring language becomes less condition-specific while the other advice-minus-factual proxy gaps stay positive.

RLVR and the final merge move no gap significantly (every dpo→rlvr and final-merge transition has $p \geq 0.168$), consistent with RLVR's task-focused math and instruction-following-constraint data (Section 4.3). These significance tests are internally valid for the frozen detector outputs; because the audit finds condition-dependent false positives and does not span checkpoints, they do not by themselves establish changes in human-judged behavior prevalence.

Axis-targeted controls qualify advice specificity. The four-seed Base/SFT/DPO extension tests 24 planned stage×control interactions (Appendix E). Four survive BH correction at $\tau = .82$. For empathy relative to emotional non-advice, the 7B Base→SFT/interface transition increases the advice gap (+0.358, $q = .016$), but DPO reverses it at both 1B (−0.467, $q = .001$) and 7B (−0.642, $q = .001$). At 7B, SFT also shifts structuring toward neutral advice (−0.208, $q = .026$). No validation or referral interaction survives $q < .05$; 1B DPO validation tendencies have $q = .059$–.095. The threshold sweep yields 3/4/5 significant inter-

actions at .80/.82/.84; only the two 7B empathy transitions survive at all three thresholds. Thus the broad advice-minus-factual changes cannot all be read as advice-specific.

4.3 The data layers provide compatible phrase-level evidence

SFT: a finite, auditable phrase-matching set (phrase-sparse). Assistant-only search of each model's exact mixture surfaces thousands of assistant turns containing referral phrases: "health-care professional" (3075/3106 for 1B/7B), "mental health professional" (2109/2120), "professional help" (1765/1774), "seek professional" (1353/1355). Length-matched recoverability is above topical controls at $\geq$ 3 words (3-word: 20.5× at 1B, 7.3× at 7B; 4-word: 136.9×/14.5×) but **phrase-sparse**: only 3/6 distinct 3-word types, dominated by the list preamble "here are some." The distinct-phrase leave-one-out is the honest figure: dropping that one phrase lowers the 3-word mean from 5455.2 to **1305.2**/M (1B) and 3026.0 to **1065.9**/M (7B) (Appendix F). Any inference must use the 3–6 distinct types as its unit, not the 45/55 occurrences. Role-scoping remains necessary: blob search inflates "talk to a" 1.41–1.43× via user turns.

DPO: paired effects, phrase sensitivity, and dependence. The exact mixes contain 378,301/378,341 preference pairs. Chosen responses are slightly *shorter* in aggregate (chosen/rejected mean-word ratio 0.990/0.997), but this aggregate normalization does not remove within-pair length as a possible covariate. At pair level, referral/disclaimer matches have absolute chosen-minus-rejected effects of 0.100 percentage points (pp; 95% CI $[.065, .135]$) at 1B and 0.179 pp ($[.144, .214]$) at 7B; matched odds ratios (MORs) are 1.181 and 1.352. Structuring effects are larger: 1.393 pp ($[1.278, 1.507]$) and 2.280 pp ($[2.167, 2.394]$). Empathy is weakly enriched only at 1B after correction; validation is not (Table 5).

Within-pair length stratification changes this interpretation. Only 19.1%/18.7% of pairs fall within ±20 words. In that comparable-length stratum, referral/disclaimer differences are −.010/+.009 pp (MOR .97/1.03, raw p = .763/.810), and high-precision referral differences are +.017/−.003 pp (p = .299/.925). Structuring remains positive but much smaller: +.184/+.343 pp (raw p = .047/.0002). In the two noncomparable strata, signs follow the longer response side: when chosen is longer by more than 20 words, referral is +.470/+.502 pp; when rejected is longer, it is −.310/−.223 pp. Structuring behaves similarly. Thus aggregate DPO enrichment is strongly length-aligned; marginal per-word ratios do not establish a within-pair preference effect.

A narrower referral lexicon that excludes the broad "consult/talk/speak to a" phrases remains positive: 0.023 pp ($[.006, .040]$, MOR 1.17, raw p = .010) at 1B and 0.043 pp ($[.026, .061]$, MOR 1.34, $p < 2 \times 10^{-6}$) at 7B. The original referral direction is positive in 8/9 and 9/9 source subsets; the high-precision direction is positive in 6/9 and 7/9. Structuring is positive in 8/9 sources at each size and is driven mainly by "first," and "here are some." Only 5.7% of rows belong to a duplicated prompt-text group (maximum multiplicity 15), so ordinary row-level McNemar inference may be mildly optimistic but is not dominated by repeated prompts. Full phrase/source results and intervals are in Appendix G. These co-occurrences are compatible with, but do not attribute or directly validate, checkpoint changes.

RLVR phrase search: near-zero counts. The RLVR datasets (GSM/MATH/IF) contain essentially none of the advice phrasings: every referral, empathy, and validation phrase counts 0–1 across `RLVR-GSM`, `RLVR-MATH`, and the IF-constraints set (only generic markers appear: "first," 881, "here are some" 48 in the IF set). We find no direct lexical evidence that the searched RLVR sets contain these proxies. Their stagewise stability (Table 4) is consistent with lexical retention rather than new introduction at RLVR.

Pretraining: not a verbatim echo (secondary, hedged). Against the exact Stage-1 corpus, advice answers are more novel than factual ones (0.556 vs 0.365 at 1B; 0.518 vs 0.371 at 7B). This is verbatim-only, length-confounded, and whole-answer rather than behavior-span, so we read it only as: advice answers are not verbatim OLMo-Mix copies. The base checkpoints' nonzero detector rates are the stronger evidence that the corresponding markers already occur at Base.

4.4 Detector audit and threshold sensitivity

A condition-stratified random audit samples 120 final-model sentences across all five conditions (40 advice, 30 factual, 20 emotional non-advice,

Model	Stage	Empathy	Validation	Referral	Structuring
1B	base	.217 (.204)	.073 (.073)	.483 (.430)	.757 (.430)
1B	sft	.393 (.393)	.167 (.167)	.697 (.684)	.847 (.767)
1B	dpo	.550 (.510)	.380 (.380)	.933 (.873)	1.000 (.400)
1B	rlvr	.567 (.507)	.410 (.410)	.923 (.836)	1.000 (.440)
1B	instruct	.590 (.550)	.363 (.356)	.900 (.820)	1.000 (.420)
7B	base	.230 (.230)	.087 (.080)	.577 (.524)	.853 (.486)
7B	sft	.493 (.493)	.183 (.183)	.767 (.767)	.787 (.714)
7B	dpo	.597 (.564)	.330 (.330)	.923 (.850)	.977 (.437)
7B	instruct	.617 (.590)	.383 (.370)	.913 (.853)	.970 (.423)

Table 4: Stagewise advice detector-positive rate with the **advice-minus-factual proxy gap in parentheses** ($\tau = 0.82$). Transition tests and corrected values appear in Table 3. Base checkpoints use plain QA; all later checkpoints use chat serialization, so Base→SFT differences conflate stage and interface.

Model	Marker proxy	Δ pp	MOR	C/R only	BH q
1B	empathy opener	.035	1.082	1768/1634	.036
1B	validation phrase	.006	1.130	183/162	.282
1B	referral/disclaimer	.100	1.181	2478/2098	$< 10^{-7}$
1B	structuring	1.393	1.242	27047/21779	$< 10^{-124}$
7B	empathy opener	.030	1.069	1734/1622	.074
7B	validation phrase	.009	1.224	191/156	.078
7B	referral/disclaimer	.179	1.352	2602/1925	$< 10^{-22}$
7B	structuring	2.280	1.435	28466/19838	$< 10^{-300}$

Table 5: Aggregate DPO pair-level effects, unadjusted for within-pair length. Δ pp is the absolute paired risk difference in percentage points; MOR is chosen-only/rejected-only. q is Benjamini–Hochberg adjusted across eight exact McNemar tests. Confidence intervals and descriptive per-word ratios are reported in the text and artifact.

20 neutral advice, 10 domain factual). The sample contains 56 1B and 64 7B sentences from 97 distinct seed-0 responses; it was not stratified on detector predictions, and multiple sentences can come from one response. Two independent blinded Codex passes agree exactly on 117/120 sentences (97.5%); category agreement is 99.2–100%, with Cohen's $\kappa = .885/.663/1.000/.978$ for empathy/validation/referral/structuring. Claude Opus 4.8 adjudicates the three disagreements. This documents AI consensus under these annotation prompts and models, not human ground truth.

Against that consensus, the union detector has **low precision and category-dependent recall** (Table 6; empathy and validation recall are each 0.50). The errors are also condition-dependent: professional referral produces 12 false positives on advice, one on emotional non-advice, two on neutral advice, and none on factual or domain-factual items; validation's two false positives are both advice items. Thus subtraction does not guarantee cancellation, and even cross-condition directions remain provisional. `crisis_referral` again fails (0 TP / 11 FP) and is excluded. The detector substantially overpredicts positives in this audit sample, suggesting that absolute output rates are upward-biased; they are not formal upper bounds because recall is below one. The stagewise results should be read as trajectories of a frozen proxy detector, not estimates of semantic behavior prevalence.

Component ablation. Exact substring matching alone produces no false positives in the 120-sentence audit but misses many positives: TP/FP/FN is 2/0/2 (empathy), 1/0/1 (validation), 1/0/0 (referral/disclaimer), and 9/0/21 (structuring). Consequently every union false positive in Table 6 is introduced by the embedding-threshold component. The union improves structuring recall from .30 to .87, but at a large precision cost. Component choice is therefore substantive, not merely a recall optimization.

Response length and fixed opportunity. The recovered 700-output final-7B cache exactly reproduces the frozen full-response rates. Advice answers average 244 tokens versus 187 for factual answers; emotional, neutral-advice, and domain-factual controls average 251–255. A truncation proxy (≥ 255 tokenizer tokens) fires for 88% of advice and 57% of factual answers, versus 96–98% of targeted controls. Limiting every response to its first 64 tokens changes the union empathy advice-minus-emotional contrast from $-.147$ to $+.253$; validation remains positive ($.250 \rightarrow .177$), referral is similar ($.090 \rightarrow .113$), and structuring remains near zero ($-.030 \rightarrow -.033$). On full answers, exact-only targeted contrasts are $.323/.153/.157/.390$ for empathy/validation/referral/structuring, respectively.

category	P	R	F1	tp/fp/fn
empathy expr.	0.22	0.50	0.31	2/7/2
validation	0.33	0.50	0.40	1/2/1
referral	0.06	1.00	0.12	1/15/0
structuring	0.49	0.87	0.63	26/27/4

Table 6: Union-detector P/R/F1 against the adjudicated 120-sentence AI consensus ($\tau = 0.82$). The two blinded AI passes agreed on 117/120 items; Opus adjudicated three. This is not human validation. Response-cluster bootstrap intervals are released with the artifact. Exact-only matching has zero false positives but low recall; `crisis_referral`: 0 TP / 11 FP, dropped.

Thus the empathy and structuring interpretations are component- or opportunity-sensitive. Excluding the 12 reused pilot prompts changes final-7B advice-minus-factual union gaps by at most .045 (Appendix D).

The low positive prevalence yields wide category-level response-cluster bootstrap intervals (Appendix H). The seed-0 sentence sample does not directly validate answer-level emission, and the audit is AI-only. We release a blinded two-human workflow but do not claim it was completed. The threshold sweep is informative rather than uniformly reassuring (Appendix C). These are robustness checks on a detector, not a validated taxonomy.

5 Discussion

From "advice template" to trained marker trajectories. The detector-positive decomposition and stagewise gaps nominate a testable account: relevant language is already detected at Base; every advice-minus-factual proxy gap increases across the confounded Base→SFT/interface transition (six of eight survive correction); across same-interface SFT→DPO, validation gaps increase, 1B referral increases, and structuring becomes less condition-specific. Axis-targeted controls show a threshold-robust 7B Base→SFT/interface increase and DPO reversal in empathy specificity. The response-level stress test then limits this account: the empathy contrast reverses under a fixed 64-token window, and exact-only versus union detectors disagree strongly about structuring specificity. Exact DPO matching removes embedding-threshold error, but not opportunity: aggregate referral/disclaimer and structuring differences largely follow whichever response is longer. Among comparable-length pairs, referral vanishes and only a small structuring difference remains. Searched RLVR data do contain near-zero affective/referral phrases. These results depend on the seeded lexicon and establish co-occurrence, not preference attribution or causation.

Safety implications. For referral/disclaimer phrasing, the bundle's safety-relevant proxy, exact search identifies a finite set of matching SFT assistant turns and measurable DPO chosen-only pair enrichment. The latter is strongly length-aligned, so it cannot be assigned specifically to preference annotations. The corpus trace still makes the marker auditable and motivates, but does not prove, a fragility hypothesis: mixture or preference-annotation changes could weaken it. The causal leave-cluster-out test our trace nominates (remove the referral cluster, re-SFT, compare against random-cluster removal) is the direct check.

Methodological lessons. (1) Occurrence-weighted provenance ratios are fragile: report distinct phrase types and a leave-one-out. (2) Advice-side rates across stages are ambiguous; report stage×condition *gaps*. (3) Whole-conversation search miscredits user turns; scope to assistant turns. (4) Cluster inference at the prompt, not the sample. (5) Separate exact and embedding components and equalize response opportunity before interpreting semantic or condition-specific contrasts.

6 Conclusion

Machine-advice outputs in OLMo-2 contain several measurable surface-marker families rather than one perfectly coupled bundle. A frozen union detector produces distinct stage and condition trajectories, but its Base→SFT changes are interface-confounded and its empathy contrast reverses under a fixed 64-token window. Exact DPO matches are more common on the chosen side in aggregate, but referral differences disappear among responses within 20 words and structuring shrinks from 1.39–2.28 pp to .18–.34 pp. The paper therefore does not establish that DPO specifically selects these markers. Searched RLVR data contain almost no affective or referral phrases, although generic structuring markers remain. The contribution is an auditable trace plus stress tests showing why detector opportunity, response length, and serialization preclude stronger claims.

7 Limitations

- **Observational throughout.** Stage localization and data co-occurrence, including the DPO chosen/rejected differential, establish no causation; the nominated leave-cluster-out re-SFT is future work.
- **Operational measurement, not semantic ground truth.** The 120-sentence AI-consensus audit finds condition-dependent false positives and category-dependent recall; exact-only matching has zero observed false positives but much lower recall. The sample comes from 97 seed-0 responses, does not validate response-level prevalence, and has no completed human pass. Threshold and component sensitivity leave all output trajectories provisional.
- **Length and opportunity.** Outputs differ sharply in length and often reach the generation cap. A 64-token window reverses the final-7B empathy contrast. Raw-output stress tests are available only for this one checkpoint, so they cannot repair or validate the stagewise trajectory.
- **Stage/interface confounding.** Base uses plain QA; SFT and later checkpoints use chat serialization. With no crossed checkpoint$\times$interface experiment, every Base$\rightarrow$SFT change is a combined stage/interface effect.
- **Controls and nulls.** The emotional, domain-factual, and neutral-advice sets target different proxy axes but are not topic-paired to the 60 advice prompts. A non-significant interaction is not evidence of equivalence.
- **Artifact and prompt scope.** The full 140-prompt battery now ships, as does the recovered final-7B raw cache; raw generations for all other checkpoints are unavailable. Twelve pilot prompts (8 advice, 4 factual) were reused in the confirmatory battery, although excluding them changes final-7B advice–factual gaps by at most .045.
- **DPO and corpus scope.** Aggregate referral effects are only .10–.18 pp under the original lexicon and .02–.04 pp under the narrower lexicon; they become approximately zero among pairs within ± 20 words. Structuring also shrinks sharply. The three-bin analysis is post hoc rather than a fully specified conditional model. Prompt-text duplication (5.7% of rows) can further make row-level inference optimistic. SFT recoverability is phrase-sparse (3–6 3-word types); Stage-2 mid-training remains unindexed; novelty is verbatim-only and length-confounded.
- **External validity.** Five canonical seeds and four extension seeds do not eliminate sampling variance. We study one model family, two sizes, one decoding configuration, and modest author-written prompt sets; no scaling or cross-family claim follows.

Ethics Statement

Data and human subjects. This work uses only publicly released artifacts (OLMo-Mix-1124 via the public infini-gram API; the OLMo-2 Tulu-3 SFT mixtures, DPO preference mixes, and RLVR sets; open-weight OLMo-2 models and their public intermediate checkpoints). We collected no new data from people. All 140 prompts are short, author-written strings describing hypothetical situations with no personal information. The study involves no human subjects and needed no ethics-board review.

Sensitive domain. Our prompts touch emotional-support, health, and relationship topics, and our analysis concerns surface phrases that can mark safety-relevant moves such as professional referrals. The audit sample shows substantial detector overprediction, suggesting upward-biased absolute rates; we also drop `crisis_referral`. Nothing here should be read as evidence that the studied models reliably provide crisis support; they were not evaluated for clinical accuracy and are not mental-health tools.

Dual use. Making the provenance of a safety-relevant marker inspectable could in principle guide its removal from a training mixture; we frame this as a risk to monitor, show co-occurrence rather than causation, and release an analysis tool, not any model or dataset that could cause direct harm.

References

Abeer Badawi, Elahe Rahimi, Md Tahmid Rahman Laskar, Sheri Grach, Lindsay Bertrand, Lames Danok, Prathiba Dhanesh, Jimmy Huang, Frank

Rudzicz, and Elham Dolatabadi. 2026. When can we trust LLMs in mental health? large-scale benchmarks for reliable LLM evaluation. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3873–3896. Association for Computational Linguistics.

Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, Usvsn Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2397–2430. PMLR.

Sreyan Ghosh, Chandra Kiran Reddy Evuru, Sonal Kumar, S. Ramaneswaran, Deepali Aneja, Zeyu Jin, Ramani Duraiswami, and Dinesh Manocha. 2024. A closer look at the limitations of instruction tuning. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Andreas Köpf, Yannic Kilcher, Dimitri von Rütte, Sotiris Anagnostidis, Zhi-Rui Tam, Keith Stevens, Abdullah Barhoum, Nguyen Minh Duc, Oliver Stanley, Richárd Nagyfi, Shahul ES, Sameer Suri, David Glushkov, Arnav Dantuluri, Andrew Maguire, Christoph Schuhmann, Huu Nguyen, and Alexander Mattick. 2023. OpenAssistant conversations – democratizing large language model alignment. In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*.

Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V. Miranda, Alisa Liu, Nouha Dziri, Shane Lyu, Yuling Gu, Saumya Malik, Victoria Graf, Jena D. Hwang, Jiangjiang Yang, Ronan Le Bras, Oyvind Tafjord, Chris Wilhelm, Luca Soldaini, and 4 others. 2024. Tulu 3: Pushing frontiers in open language model post-training. *Preprint*, arXiv:2411.15124.

Jiacheng Liu, Taylor Blanton, Yanai Elazar, Sewon Min, Yen-Sung Chen, Arnavi Chheda-Kothary, Huy Tran, Byron Bischoff, Eric Marsh, Michael Schmitz, Cassidy Trier, Aaron Sarnat, Jenna James, Jon Borchardt, Bailey Kuehl, Evie Yu-Yen Cheng, Karen Farley, Taira Anderson, David Albright, and 11 others. 2025. OLMoTrace: Tracing language model outputs back to trillions of training tokens. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL): System Demonstrations*, pages 178–188, Vienna, Austria. Association for Computational Linguistics.

Jiacheng Liu, Sewon Min, Luke Zettlemoyer, Yejin Choi, and Hannaneh Hajishirzi. 2024. Infini-gram: Scaling unbounded n-gram language models to a trillion tokens. In *First Conference on Language Modeling (COLM)*.

Siyang Liu, Chujie Zheng, Orianna Demasi, Sahand Sabour, Yu Li, Zhou Yu, Yong Jiang, and Minlie Huang. 2021. Towards emotional support dialog systems. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3469–3483. Association for Computational Linguistics.

Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912. Association for Computational Linguistics.

Pamela D. Rivière, Cameron Jones, and Sean Trott. 2026. Developmental trajectories of situation modeling and mentalizing in transformer language models. *Preprint*, arXiv:2606.28524.

Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, and 24 others. 2025. 2 OLMo 2 Furious. *Preprint*, arXiv:2501.00656. A shorter version appeared at COLM 2025.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual e5 text embeddings: A technical report. *Preprint*, arXiv:2402.05672.

Anuradha Welivita, Fawzia Zeitoun, and Pearl Pu. 2026. Who generates more empathetic responses—humans or LLMs? a comparative evaluation with human and LLM judges. In *Proceedings of the 30th Conference on Computational Natural Language Learning*, pages 358–381. Association for Computational Linguistics.

Mengzhou Xia, Mikel Artetxe, Chunting Zhou, Xi Victoria Lin, Ramakanth Pasunuru, Danqi Chen, Luke Zettlemoyer, and Veselin Stoyanov. 2023. Training trajectories of language models across scales. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13711–13738. Association for Computational Linguistics.

Mengzhou Xia, Sadhika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. 2024. LESS: Selecting influential data for targeted instruction tuning. In *Proceedings of the 41st International Conference on Machine Learning*.

Frank Xiao and Santiago Aranguri. 2026. Probe-based data attribution: Discovering and mitigating undesirable behaviors in LLM post-training. *Preprint*, arXiv:2602.11079. ArXiv preprint, v3.

Chunting Zhou, Pengfei Liu, Puxin Xu, Srini Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu,

Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. 2023. LIMA: Less is more for alignment. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.

A Prompts and decoding configuration

All answers use nucleus sampling (temperature 0.7, top-p 0.9, `max_new_tokens`=256, `do_sample=True`), the chat template where available (plain `Question:/Answer:` for base checkpoints), and five seeds per prompt in the canonical grid. The battery: 60 advice, 30 factual, 20 emotional non-advice, 20 neutral advice, 10 domain factual; the complete machine-readable list ships as `experiments/prompts_full.json`.
The Base/SFT/DPO axis-targeted-control extension uses four seeds; all balanced extension tests use seeds 0–3 for every condition. Eight advice and four factual prompts were reused from the earlier 12-prompt pilot; the other 128 are new to the full battery.

Examples.

- Advice: `I've been feeling really anxious lately and can't sleep. Any advice?`
- Factual: `How does photosynthesis work?`
- Emotional non-advice: `Describe the common symptoms of loneliness.`
- Neutral advice: `How should I organize my bookshelf?`
- Domain factual: `What are common clinical treatments for insomnia?`

B Surface-marker lexicon and detector threshold

A category fires for an answer if *either* (i) any lexicon phrase occurs as a case-insensitive substring, *or* (ii) any sentence has cosine similarity $\geq \tau$ to a seed exemplar (`intfloat/multilingual-e5-small`, `query:`/`passage:` prefixes), $\tau = 0.82$ primary.

Lexicon (substring match).

- **empathy-opener proxy** (`empathy_opener`): `i'm sorry to hear, i am sorry to hear, i'm really sorry, that sounds really, that sounds very, i can understand, i can imagine, it's understandable, that must be, i'm here for you, i hear you, it sounds like you`
- **validation-phrase proxy**: `your feelings are valid, it's okay to feel, it's normal to feel, it's completely normal, what you're feeling, there's nothing wrong with, it's natural to`
- **referral/disclaimer proxy** (`disclaimer`): `consult a, talk to a, speak with a, see a doctor, seek medical, i am not a doctor, i'm not a doctor, medical professional, healthcare professional, mental health professional, seek professional, professional help`
- **crisis_referral** (dropped after 0 TP / 11 FP in the 120-sentence audit): `crisis, hotline, helpline, 988, call 911, emergency services, suicide`
- **structuring-marker proxy** (`structure`): `here are some, here are a few, consider the following, steps you can take, a few things you can, first,, secondly,`

e5 seed exemplars. Empathy: "I'm so sorry you're going through this"; "That sounds incredibly hard." Validation: "Your feelings are completely valid"; "It's totally normal to feel this way." Referral: "You should consult a qualified professional"; "Please talk to your doctor about this." Structuring: "Here are a few steps you can take"; "Consider the following suggestions."

C Threshold sweep

Table 7 reports the exact contrasts underlying the decomposition at $\tau = 0.80/0.82/0.84$. Structuring is stable: advice and neutral-advice remain within .05 in every cell. Validation exceeds its strongest control in all six cells. Empathy's emotional-minus-advice contrast reverses at .84, while referral's advice-minus-domain contrast changes magnitude. The primary $\tau = .82$ was frozen before the 140-prompt battery, but its qual-

itative decomposition should not be generalized across detector thresholds.

Model	Contrast	.80	.82	.84
1B	empathy: emot.−adv.	.030	.200	−.107
7B		.030	.143	−.280
1B	validation: adv.−max ctrl.	.030	.243	.107
7B		.047	.253	.170
1B	referral: adv.−domain	−.003	.080	.093
7B		.017	.093	.280
1B	structuring: adv.−neutral	.000	.000	.047
7B		.000	−.030	−.010

Table 7: Threshold robustness of the four decomposition contrasts. Positive values support the primary-threshold reading except for structuring, where values near zero indicate the same rate on advice and neutral advice.

D Final-7B response-level robustness

The recovered `final_7b_answers.jsonl` contains all 700 final-7B responses (140 prompts $\times$ five seeds). Table 8 shows the opportunity imbalance. "Cap proxy" means at least 255 tokens under the released OLMo tokenizer; it indicates likely truncation but is not a direct generation-stop record.

Condition	n	mean tok.	mean sent.	cap proxy
advice	300	243.8	13.5	.880
factual	150	187.4	7.8	.573
emotional	100	253.5	10.9	.980
neutral advice	100	254.6	13.8	.970
domain factual	50	250.5	10.2	.960

Table 8: Final-7B response length and generation-cap proxy.

Table 9 recomputes each axis-targeted contrast using exact substring matching alone and the exact-or-embedding union, over full answers and fixed prefix windows. The full-union column exactly reproduces the frozen final-7B result. The first-64-token empathy reversal shows that its full contrast is not opportunity-robust. Exact-only matching instead makes structuring look advice-specific relative to neutral advice; the embedding component removes that contrast.

Proxy contrast	Exact full	Union full	Union 64	Union 128
empathy: adv.−emot.	.323	−.147	.253	.063
validation: adv.−emot.	.153	.250	.177	.210
referral: adv.−domain	.157	.090	.113	.117
structuring: adv.−neutral	.390	−.030	−.033	−.030

Table 9: Detector-component and fixed-window sensitivity at final 7B ($\tau = .82$ for union columns).

Among advice responses, union bundle sizes are 1/2/3/4 proxies in 4.7/29.0/40.7/25.7% of cases. Pairwise phi correlations range from $-.14$ to .21; referral/disclaimer appears in 93.0% of validation-positive responses. Excluding the 8 advice and 4 factual pilot prompts changes full-union advice-minus-factual gaps from .587/.367/.850/.423 to .542/.369/.831/.458 (empathy/validation/referral/structuring).

E Axis-targeted-control checkpoint interactions

Table 10 reports every planned interaction in the four-seed Base/SFT/DPO extension. Columns Base, SFT, and DPO are advice-minus-control gaps; transition columns are changes in those gaps with BH-adjusted q in parentheses. The family contains all 24 displayed transition tests.

F Per-phrase recoverability and leave-one-out

Per-phrase assistant-only counts against each model's exact mixture (`-0225`, 866,138 docs, 1B; `tulu-3-sft-olmo-2-mixture`, 939,344, 7B); `infl` = blob/assistant.

Length-matched buckets (occurrence-weighted). Seeded-marker vs topical per-million for 1B: 2-word 1884.3 vs 1760.2 (1.1×, n_{occ}=14), 3-word 12900.2 vs 629.8 (20.5×, 45), 4-word 1596.9 vs 11.7 (136.9×, 30). For 7B, 2-word 1784.5 vs 996.8 (1.8×, 20), 3-word 9354.4 vs 1280.4 (7.3×, 55), 4-word 1396.6 vs 96.2 (14.5×, 24). **Distinct-phrase leave-one-out (3-word):** dropping "here are some" moves the mean from 5455.2 to 1305.2/M at 1B (3 types) and from 3026.0 to 1065.9/M at 7B (6 types).

G DPO phrase and source robustness

Table 12 gives paired risk-difference and matched-odds-ratio intervals. The higher-precision referral set requires an explicit doctor or professional object (e.g., "talk to your doctor," "consult a healthcare professional," "seek professional help") and excludes broad fragments such as "consult a" and "talk to a." Intervals use the discordant-pair normal approximation for the risk difference and log-MOR; exact McNemar p-values are released.

Table 13 is the requested within-pair length diagnostic. The near-antisymmetric signs in the longer-response strata show that the aggregate chosen-side

Model	Marker	Control	Base	SFT	DPO	Base→SFT	SFT→DPO
1B	referral	domain factual	.188	.308	.167	+.121 (.414)	−.142 (.296)
1B	empathy	emotional	.017	.225	−.242	+.208 (.095)	−.467 (.001)
1B	structuring	neutral advice	−.013	−.050	.000	−.037 (.653)	+.050 (.475)
1B	validation	domain factual	.071	.146	.375	+.075 (.475)	+.229 (.095)
1B	validation	emotional	.021	.121	.225	+.100 (.293)	+.104 (.296)
1B	validation	neutral advice	.058	.171	.350	+.113 (.189)	+.179 (.059)
7B	referral	domain factual	.321	.067	.200	−.254 (.166)	+.133 (.293)
7B	empathy	emotional	.108	.467	−.175	+.358 (.016)	−.642 (.001)
7B	structuring	neutral advice	.050	−.158	−.025	−.208 (.026)	+.133 (.095)
7B	validation	domain factual	.092	.175	.317	+.083 (.475)	+.142 (.296)
7B	validation	emotional	.054	.125	.179	+.071 (.414)	+.054 (.537)
7B	validation	neutral advice	.092	.163	.292	+.071 (.414)	+.129 (.189)

Table 10: Complete axis-targeted-control stage interactions at $\tau = .82$ (10k prompt-clustered permutations). Four of 24 transition tests have $q < .05$. Only the 7B empathy Base→SFT and SFT→DPO effects retain $q < .05$ at all three detector thresholds.

Model	Phrase	Asst.	Blob	Infl.
1B	first,	44422	52676	1.19
1B	here are some	11914	13276	1.11
1B	here are a few	5421	6056	1.12
1B	healthcare professional	3075	3216	1.05
1B	mental health professional	2109	2186	1.04
1B	professional help	1765	1825	1.03
1B	seek professional	1353	1377	1.02
1B	it sounds like you	1130	1164	1.03
1B	consult a	891	978	1.10
1B	it's understandable	218	227	1.04
1B	it's natural to	152	157	1.03
7B	here are some	12049	13432	1.11
7B	here are a few	5511	6156	1.12
7B	healthcare professional	3106	3309	1.07
7B	i'm really sorry	2369	2378	1.00
7B	mental health professional	2120	2214	1.04
7B	professional help	1774	1838	1.04
7B	seek professional	1355	1379	1.02
7B	it sounds like you	1139	1184	1.04
7B	consult a	927	1031	1.11
7B	talk to a	286	408	1.43
7B	it's completely normal	78	103	1.32

Table 11: Per-phrase SFT-assistant recoverability (representative rows). The 3 (1B) / 6 (7B) distinct 3-word types drive Section 4.3; "here are some" dominates the 3-word mean.

effect is strongly associated with relative response length. The comparable-length band is descriptive and post hoc; it is not a substitute for a pre-specified conditional model.

For the original referral group, positive chosen-minus-rejected differences occur in 8/9 1B sources (one tie) and all 9/9 7B sources. Under the higher-precision group they occur in 6/9 and 7/9 sources. Structuring is positive in 8/9 sources at each size; the negative source is the instruction-following personas subset. The main structuring contribution comes from "first," (.931/1.381 pp at 1B/7B) and "here are some" (.339/.736 pp), rather than every phrase moving uniformly. The main referral contributors are "professional help," "consult a," and named professional phrases. There are 367,082/367,110 distinct prompt texts; 21,551/21,574 rows (5.7%) belong to duplicated groups, with maximum multiplicity 15.

H Response-cluster audit uncertainty

The 120 audit sentences come from 97 seed-0 responses. To preserve dependence among sentences from the same response, we assign Dirichlet(1) weights to response keys in a 10,000-draw Bayesian bootstrap and recompute each metric. Table 14 reports percentile intervals. These intervals quantify sampling uncertainty around the AI consensus labels; they do not turn the audit into human validation.

I Statistical formulas

Primary inference uses two-sided prompt-clustered permutation tests on prompt-level means (10,000 draws), with Monte Carlo $p = (b + 1)/(B + 1)$; $p < 10^{-4}$ denotes zero raw exceedances. The distinct-phrase leave-one-out removes the most frequent 3-word phrase before recomputing the mean seeded-marker per-million. DPO comparisons use exact McNemar tests on chosen-only versus rejected-only pairs, Haldane–Anscombe matched odds ratios with log-scale intervals, paired risk differences with normal-approximation intervals, and phrase occurrences per million whitespace-delimited response words. Released scripts derive all statistics from the artifact.

Model	Phrase group	Δ pp [95% CI]	MOR [95% CI]	C-only	R-only
1B	referral/disclaimer	.100 [.065,.135]	1.181 [1.114,1.252]	2478	2098
1B	high-precision referral	.023 [.006,.040]	1.170 [1.039,1.318]	591	505
1B	structuring	1.393 [1.278,1.507]	1.242 [1.220,1.264]	27047	21779
7B	referral/disclaimer	.179 [.144,.214]	1.352 [1.274,1.434]	2602	1925
7B	high-precision referral	.043 [.026,.061]	1.335 [1.187,1.502]	649	486
7B	structuring	2.280 [2.167,2.394]	1.435 [1.409,1.461]	28466	19838

Table 12: DPO paired absolute and relative effects. Positive Δ means more chosen-only than rejected-only matches.

Model	Word-difference band	n	Referral		High precision		Structuring	
			Δ pp	MOR	Δ pp	MOR	Δ pp	MOR
1B	chosen longer > 20	171170	+.470	1.991	+.059	1.408	+5.979	2.361
1B	rejected longer > 20	134707	$-.310$	.616	$-.020$	.871	-3.786	.579
1B	within ± 20	72424	$-.010$	.965	+.017	1.238	+.184	1.062
7B	chosen longer > 20	187201	+.502	2.172	+.098	1.761	+6.562	2.610
7B	rejected longer > 20	120381	$-.223$	.706	$-.015$	.905	-3.239	.619
7B	within ± 20	70759	+.009	1.028	$-.003$	.966	+.343	1.123

Table 13: DPO effects stratified by chosen-minus-rejected response length. Positive Δ means more chosen-only matches.

Category	Precision 95%	Recall 95%	F1 95%
empathy expression	[.04,.46]	[.10,.90]	[.05,.57]
validation	[.02,.84]	[.03,.98]	[.02,.83]
referral	[.00,.22]	[1.00,1.00]	[.00,.36]
structuring	[.35,.63]	[.74,.96]	[.49,.74]

Table 14: Bayesian response-cluster bootstrap intervals for the 120-sentence AI consensus audit. Referral recall is degenerate because the single positive gold item is detected in every draw; its precision and F1 remain highly uncertain.